SOFTWARE CONSTRUCTION & DESIGN

Lightweight Project Requirements Document

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IM SCIENCES

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Project Title: Chicken Stock — Personal Stock Management System

1. Description / Problem Statement

Investors often need to check multiple sources to monitor stock prices, track their investments, and calculate their profits or losses. Managing this information manually can be inconvenient and make it difficult to understand overall portfolio performance.

Chicken Stock aims to provide a centralized application that uses real-world stock market data to help users manage their holdings and monitor investment performance.

2. Project Objectives

- Retrieve stock market data through an external API.
- Allow users to record and manage their stock holdings and transactions.
- Automatically calculate investment value, profit/loss, and returns.
- Visualize stock price movements and portfolio performance.
- Provide a simple, user-friendly interface for monitoring investments.

3. Intended Users

- Individual investors who want to track their investments.
- Beginner investors who want a clear overview of their stock holdings.
- Users interested in monitoring stock prices and market performance.

4. Functional Requirements

ID	Requirement
FR-01	The system shall retrieve stock prices and available market information through an external API.
FR-02	Users shall be able to search for stocks using their names or ticker symbols.
FR-03	Users shall be able to add stocks to their personal portfolio by entering the quantity and purchase price.
FR-04	Users shall be able to record buy and sell transactions and update their holdings accordingly.
FR-05	The system shall calculate total investment, current portfolio value, and profit/loss.
FR-06	The system shall display stock price changes and historical price charts when data is available.
FR-07	The system shall display portfolio allocation by stock and, where data permits, by sector.
FR-08	Users shall be able to view, edit, and remove their portfolio entries.
FR-09	The system shall handle unavailable market data and API errors by displaying appropriate messages.

5. Non-Functional Requirements

- **Usability:** The interface should be simple and understandable for beginner investors.
- **Performance:** Common actions and portfolio calculations should complete promptly under

normal conditions.

- **Reliability:** The application should handle invalid inputs, network failures, and API rate limits gracefully.
- **Security:** API keys must not be exposed in public client-side code. User data should be protected against unauthorized access.
- **Maintainability:** The code should be modular, with market-data retrieval, portfolio calculations, and interface logic separated.
- **Accuracy:** Financial calculations should use consistent precision and be based on the latest available market data, with timestamps or delay information shown where possible.

6. Prioritized Product Backlog

Priority	Feature	Description
P1 — Must have	Market data integration	Retrieve real-world stock prices through an API.
P1 — Must have	Portfolio management	Add and manage stock holdings.
P1 — Must have	Financial calculations	Calculate investment value and profit/loss.
P1 — Must have	Transaction management	Record purchases and sales.
P2 — Should have	Historical charts	Display stock price movements over time.
P2 — Should have	Portfolio dashboard	Summarize holdings, returns, and total portfolio value.
P2 — Should have	Allocation visualization	Show how investments are distributed across stocks and sectors.
P3 — Could have	Watchlist	Track stocks without owning them.
P3 — Could have	Price alerts	Notify users when a stock reaches a specified price, if feasible.
P3 — Could have	User accounts	Support separate portfolios for different users, if time permits.

7. Initial Assumptions and Constraints

Assumptions

- The application will initially focus on a single supported stock market or a limited set of markets.
- Users will enter their own purchase prices and quantities.
- A stock-data API will provide the required information under its available access terms.
- Market data will be used for informational purposes rather than personalized investment advice.

Constraints

- The project must be completed within the course timeline and available group resources.
- Free API plans may impose request limits or provide delayed rather than real-time prices.
- The application depends on internet connectivity to retrieve updated market data.
- Advanced features such as automated trading, price notifications, and multi-user support

may be excluded from the initial version.

Known Limitations

- Displayed prices may be delayed and may not reflect executable trading prices.
- Taxes, brokerage fees, dividends, and currency conversion may not be included in the initial profit/loss calculations.
- Portfolio performance calculations will depend on the accuracy of user-entered transactions and the availability of historical market data.

8. Agile Development Approach

Development will proceed in small iterations. The group will first implement market-data retrieval, portfolio management, and financial calculations. Subsequent iterations will focus on charts, visualizations, testing, and usability improvements.

Requirements and priorities may change based on development progress, API limitations, testing results, and instructor feedback. At the end of each iteration, the group will review completed features and refine the backlog for the next iteration.

Expected initial deliverable: A working application that retrieves real-world stock prices, allows users to record their holdings, and calculates and displays portfolio performance.